A16728W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B4: NUCLEAR AND PARTICLE PHYSICS

TRINITY TERM 2021

Tuesday, 08 June

Opening time: 9.30 am UK Time

You have two hours writing time to complete the paper and up to 30 minutes technical time to upload your answer file.

Answer **two** *questions.*

Write your candidate number, paper name, paper code and a list of the questions that you have answered on the first page, and submit all of your answers as a single PDF file.

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors will be available within Inspera during the examination

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

1. (a) Discuss the nature of and evidence for colour and flavour conservation in strong, electromagnetic and weak interactions. [5]

(b) In an electron-positron collider with equal beam energies, the following interaction is observed:

$$e^+ + e^- \to \mu^+ + e^- + \text{missing energy}.$$

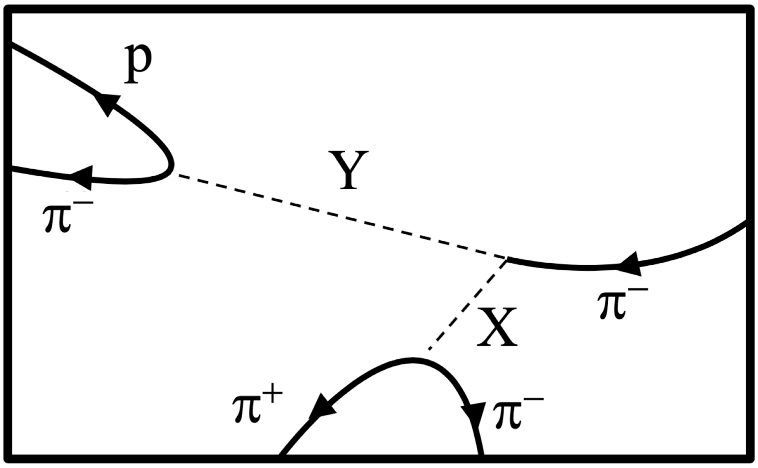
Provide two different Standard Model explanations and draw the appropriate Feynman diagrams to describe this observation. Sketch the distribution of missing energy for each case when the observed rate is maximum. [10]

(c) In a future μ^+-μ^- collider, the resonant production of tau particles is identified through the observation of two high-energy EM showers fully contained in the electromagnetic calorimeters. What fraction of the produced tau particles are observed through this channel? [4]

(d) At energies a few MeV below the resonance, what is the ratio of e^+e^- pairs produced relative to the total number of interactions seen to produce hadrons? [6]

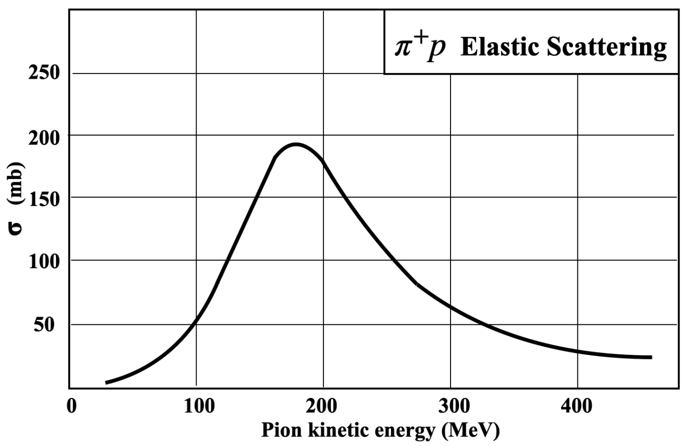
2. (a) List and briefly discuss the various conserved quantum numbers for the strong, electromagnetic, and weak nuclear forces in the Standard Model. [3]

(b) The figure below represents tracks observed when an incoming π^-, entering on the right side of the figure, interacts with a proton inside the detector. Charged particle tracks are shown by the solid lines, whereas the dashed lines labelled X and Y represent the trajectories of unobserved neutral particles that are inferred. Assume the tracks all lie in the plane of the paper and that a magnetic field is directed into that plane. Calorimetric measurements are made at the terminus of each track that leaves the area of the box. Explain how the identities of the produced charged particles can therefore be determined. [2]



(c) From the use of conservation laws, and the knowledge that $\Sigma^0 \to \Lambda + \gamma$, deduce the identities of the mystery particles X and Y. [10]

(d) The figure below shows the measured elastic cross section for $\pi^+ p \to \pi^+ p$ as a function of incoming pion kinetic energy for a fixed target experiment. Estimate the rest mass and lifetime of the resonant particle state produced. [3]



(e) Estimate the expected non-relativistic form of the cross section vs pion energy and draw this on the same plot. Suggest an explanation for any observed deviations. [7]

3. (a) Define the scattering amplitude in the context of the first-order Born approximation and explain the validity range of this approximation. [3]

(b) Taking the incoming wave to be along the z-direction ($\mathbf{k} = k\hat{\mathbf{z}}$), use the first order Born approximation to show that the forward scattering amplitude for scattering from a Gaussian potential:

$$U(r) = U_0 \mathrm{e}^{-r^2/2\sigma^2}$$

has the form:

$$f(\theta, k) = A \exp[-\sigma^2 k^2 (1 - \cos\theta)]$$

where θ is the scattering angle relative to the direction of the incoming wave. Determine the factor A. [6]

(c) Show that the total cross section has the form:

$$\sigma_{\mathrm{tot}}(k) = B\left(1 - \mathrm{e}^{-4\sigma^2 k^2}\right)$$

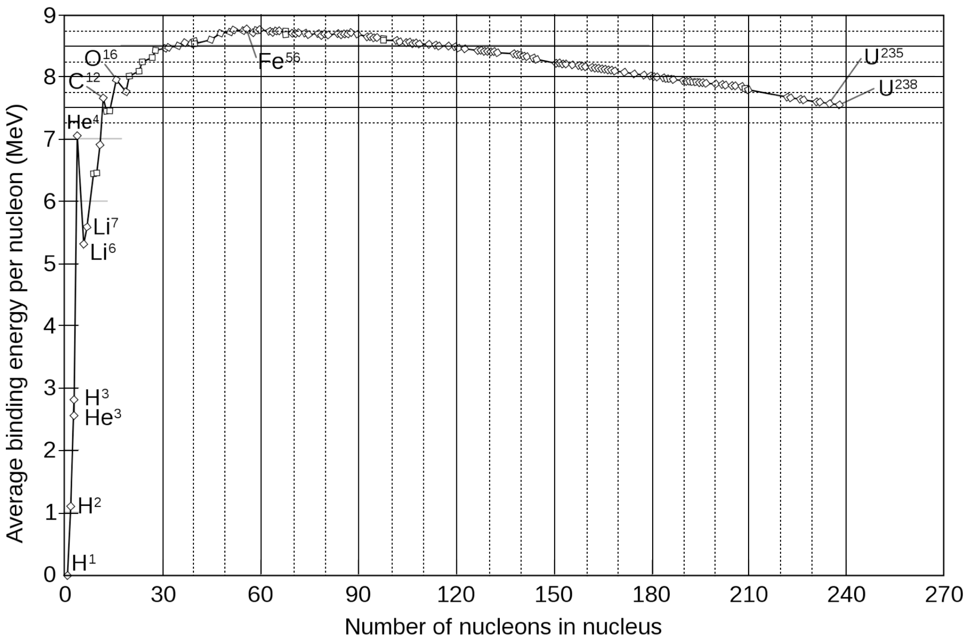
and determine the factor B. [6]

(d) For energies below about 50 MeV, such a Gaussian form can be used to model the nucleon-nucleon scattering potential. Assume σ is of the order of the nucleon radius, $\sim$1 fm. Estimate the intrinsic momenta of nucleons bound to each other on this scale. [2]

(e) Write an approximate expression for the total energy of the deuteron, a bound state of n + p. Given that the deuteron can be broken apart by radiation with energy of 2.2 MeV or more, estimate a value for U_0. [4]

(f) From this, estimate the mean free path for a 10 MeV neutron in water, assuming scattering is dominated by interactions with hydrogen. [4]

4. (a) The figure below shows the binding energy per nucleon as a function of atomic number. Discuss the factors that determine the characteristic features of this curve. Include a description of the terms in the semi-empirical mass formula, the reasons for their signs, and the range of validity for this approximation. [5]



(b) A typical reaction in a nuclear reactor is:

$$n + {}^{235}_{92}\mathrm{U} \rightarrow {}^{236}_{92}\mathrm{U} \rightarrow {}^{92}_{36}\mathrm{Kr} + {}^{141}_{56}\mathrm{Ba} + 3n.$$

Estimate how much energy is released per fission in units of MeV. [5]

(c) Assume the produced daughter isotopes undergo beta-decay to stable isotopes. Estimate the rate of anti-neutrinos produced by a nuclear reactor with a 10 GW thermal power. [3]

(d) We wish to monitor the operation of such a reactor from a distance of 200 km by measuring inverse beta-decay (IBD) interactions in a liquid scintillator detector with a composition of $C_{20}H_{34}$. Assume that 90% of anti-neutrinos fall below the kinetic threshold energy of 1.8 MeV for the IBD reaction, but that the detector is otherwise 100% efficient in observing the reaction. A typical anti-neutrino inverse beta-decay cross section of hydrogen for reactor energies is $\sim 4 \times 10^{-42}\,\mathrm{cm}^2$. Estimate the detector mass required to observe 10 events in a month. [10]

(e) In practice, only about half this number of anti-neutrino interactions are observed in such a detector. What assumption has been left out in the above calculation that could account for this? [2]